MASKINHALLEN · USER MANUAL · VERSION 5.1

ARMERING

WAVETABLE SYNTH · EUCLIDEAN FILTER · SEQUENCER

A synth for minimal and percussive techno: two wavetable oscillators, skewed filters, euclidean rhythms that play the filter, and a sequencer with locks.



VST3 · AU · LV2 · STANDALONE
MACOS · WINDOWS · LINUX

MASKINHALLEN

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01 WELCOME

ARMERING is a playful wavetable synth. You play it with MIDI. It moves by itself.

Every sound has two wavetable oscillators, a crooked analog-style filter, four LFOs, three envelopes and three euclidean rhythm lanes. Anything can modulate almost anything: drag a source onto a knob and it starts to move.

- 12 wavetables with 32 frames each, and 6 ways to bend them
- 5 filters: ladder, diode, a dirty MS-20 style, a morphing state-variable and a comb
- 3 euclidean rhythm lanes with different lengths, for polyrhythm on the filter
- **MAGI**: the rhythm patterns rotate and change a little in every bar
- A 16-slot mod matrix with drag and drop, and colored rings on the knobs
- 1000 sounds in 8 categories, a dice on every section, and **SPARKA**
- Distortion, chorus, ping-pong echo and room, all ducked so the sound stays clear

ABOUT THE SWEDISH LABELS

ARMERING was built in Stockholm, so the controls have Swedish names. This manual writes every label as it looks on screen. The words you will see most:

ON SCREEN	ENGLISH	ON SCREEN	ENGLISH
`LJUD`	Sound	`KATEGORI`	Category
`ALLT`	Everything	`SPARKA`	Kick (new movement)
`Ångra`	Undo	`KLOCKA`	Clock
`Varp`	Warp	`Nivå`	Level
`Halvton` / `Fin`	Semitone / Fine	`Bredd`	Width
`Värme`	Warmth	`Key-följ`	Key tracking
`Steg` / `Slag`	Steps / Hits	`Takt`	Rate
`Källa` / `Mål`	Source / Target	`Mängd`	Amount
`Synk` / `Omstart`	Sync / Retrigger	`Mjukhet`	Smoothing
`Eko` / `Rum`	Echo / Room	`Glid`	Glide

THE WINDOW



02 INSTALL

Unzip the file for your system. You get a folder called **ARMERING**. Then run the installer for your system and restart your DAW (or rescan plug-ins). ARMERING shows up as an instrument.

MACOS

1. Double-click `ARMERING-macOS.zip` in Downloads.
2. Open Terminal and run:
   `bash ~/Downloads/ARMERING/installera.command`
3. Quit your DAW completely and start it again.

VST3, AU and a standalone app. macOS 11 or newer, Intel or Apple Silicon. The installer copies the plug-ins and removes macOS's block on downloaded plug-ins.

WINDOWS

1. Right-click `ARMERING-Windows.zip` and choose **Extract all**.
2. Open the `ARMERING` folder and right-click `installera.bat` → **Run as administrator**.
3. If Windows says "Windows protected your PC": **More info** → **Run anyway**.

VST3 (in C:\Program Files\Common Files\VST3) and a standalone .exe. 64-bit Windows 10 or 11.

LINUX

1. Unzip `ARMERING-Linux.zip`.
2. Run `bash ~/Downloads/ARMERING/installera-linux.sh`. It copies the plug-ins to `~/.vst3` and `~/.lv2`.
3. Rescan plug-ins in your DAW (Bitwig, Reaper, Ardour and others).

VST3 and LV2, 64-bit x86 Linux (built on Ubuntu 22.04).

> SWEDISH OR ENGLISH
>
> The machines have Swedish names on the panel. One click on `SV | EN` in the header switches labels, menus and tooltips to English. Your choice is remembered for every MASKINHALLEN machine.

03 UNLOCK WITH YOUR KEY

Your purchase comes with a licence key: four groups of four characters, like `XXXX-XXXX-XXXX-XXXX`. You find it in the Payhip email and on your receipt.

1. Open ARMERING. Until it is unlocked, the window shows **UNLOCK ARMERING**.
2. Paste the whole key in the field and press `UNLOCK`.

3. Done. You only need to do this once per computer: every MASKINHALLEN machine on the computer finds the key. A bundle key unlocks every machine.

The key is not case sensitive, and I/L and O/0 are treated as the same character, so a typo there does not matter. If it says the key does not work, check that the whole key was pasted. You can use your key on your own computers.

THE FREE DEMO

Every machine has a free demo with the same sound engine. Some presets and functions are locked, and the demo has its own name (ARMERING DEMO), so it can sit next to the full version. Buying the full version is what unlocks everything; install it and paste your key.

UPDATES

Updates are free for life. Download the new zip from your Payhip receipt link and run the installer again. Your projects and your key are kept.

04 YOUR FIRST SOUND IN ONE MINUTE

1. Put ARMERING on an instrument track and press Play in your DAW.
2. Pick a `KATEGORI`, for example **BAS**, and play a few low notes.
3. Step through sounds with the arrows `<` and `>`.
4. Found one you like? Press `SPARKA` a few times. The sound stays the same, but the movement changes.
5. Turn `Makro 1` to open the filter. Turn the others and listen.
6. Went too far? Press `Ångra`.

05 THE WINDOW

`Top bar`	Category, sound, dice, SPARKA, undo, 4 macros, tempo and clock.
`OSC A / OSC B`	The two wavetable oscillators. OSC B also holds Sub and Brus (noise).
`FILTER`	Five filter types, a curve that shows the shape, and six knobs.
`FILTERRYTM`	Three euclidean rhythm lanes and MAGI.
`LFO`	Four LFOs. Pick one with the tabs 1–4.
`MOD-MATRIS`	All sources as tiles, and 16 modulation slots.

ENV	Three envelopes, mono and glide.
FX	Distortion, chorus, echo, room and volume.

06 SOUNDS, DICE AND SPARKA

ARMERING has 1000 sounds in 8 categories, 125 in each:

BAS	Basslines, from round subs to gnarly wavetable basses.
LEAD	Lines that sing, sweep and bend.
PLOCK	Short plucks and bells.
MATTA	Pads and slow carpets of sound.
SEKVENS	Hold one note and the sound plays a pattern by itself.
STAB	Short chords and hits.
MASKIN	Drones, engines and machine noise that moves.
KLONK	Percussive metal, wood and glass.

THE DICE

ALLT rolls a whole new sound. Every section also has its own small dice in the top right corner: OSC A, OSC B, FILTER, FILTERRYTM, LFO, MOD-MATRIS and FX. It only changes that section and leaves the rest alone. When you change a sound, its name gets a *.

SPARKA

SPARKA keeps the oscillators, the filter and the effects, but gives the sound brand new movement: new LFOs, new rhythm lanes and new modulation. Press it until the sound starts to dance.

> UNDO
> Ångra goes back up to 30 steps: sound changes, dice rolls, SPARKA and new modulations.

07 THE OSCILLATORS

Each oscillator plays a **wavetable**: a row of 32 waveforms. Position moves through them. When an LFO or a rhythm moves the position, the tone changes all the time.

Analog	Saw to square to pulse.	Vokal	Vowels, from oo to ee.

`FM`	Soft to metallic FM.	`Metall`	Bells and clangs.
`Betong`	Dark and heavy.	`Additiv`	Harmonics that come and go.
`Sync`	Hard sync sweeps.	`Vikt`	Folded, buzzy waves.
`Orgel`	Organ drawbars.	`Glas`	Clear and glassy.
`Kör`	Choir-like formants.	`Bit`	Lo-fi and crunchy.

`Varp` bends the wave in one of six ways: **Böj** (bend), **Sync**, **Spegel** (mirror), **FM**, **Vik** (fold) and **Kross** (crush). The knob `Varp` sets how much.

`Unison` stacks up to 8 voices. `Detune` spreads them, `Blend` mixes the detuned voices in and `Bredd` spreads them in stereo.

08 THE FILTER

`Stege`	Ladder. Round and fat.
`Diod`	Diode. Sour and rubbery, with a thin low end.
`Skrik`	Scream. Dirty and sharp, in the style of an MS-20.
`Svep`	Sweep. Glides from lowpass through notch to highpass.
`Kam`	Comb. Metallic tones that follow the note.

`Drive` pushes the filter into distortion. `Key-följ` opens the filter on higher notes. `Env 2` lets envelope 2 open or close the filter on every note. `Värme` rolls off the harsh top end.

09 FILTERRYTM: EUCLIDEAN RHYTHM AND MAGI

A euclidean rhythm spreads a number of hits as evenly as possible over a number of steps. 3 hits over 8 steps gives the classic tresillo. Each hit gives a short pulse that fades out.

ARMERING has three lanes: **Rytm 1**, **Rytm 2** and **Rytm 3**. Give them different lengths, like 16, 7 and 5 steps, and they slide against each other. That is polyrhythm.

`Steg`	Length of the lane, 1–32 steps.
`Slag`	How many hits.
`Rotation`	Moves the pattern forward.

`Takt`	Step rate: 1/16, 1/8, 1/16 triplet, 1/32 or 1/8 triplet.
`Decay`	How fast each pulse fades.

To make a lane play the filter, drag its tile (`Rytm 1`) onto `Cutoff`. Many sounds already do this.

MAGI

`MAGI` makes the rhythm change a little in every bar: a lane can rotate, or get one hit more or less. At low settings it happens now and then. At high settings it happens almost every bar. The changes always land in time, and the same bar always does the same thing, so it sounds alive but never random. The dice under MAGI sets a new amount and new rotations.

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10 MODULATION

DRAG AND DROP

1. Find a source tile in the MOD-MATRIS, for example `LFO 1`.
2. Drag it onto a knob, for example `Position` in OSC A.
3. A colored ring shows up around the knob. It shows how far the modulation goes.
4. Drop the same source on the same knob again to make the modulation deeper.

Knobs that take modulation: Position, Varp, Nivå, Halvton and Fin on the oscillators, Sub, Brus, Cutoff, Resonans, Drive and Volym.

THE MOD MATRIX

The 16 slots show every modulation: **source**, **target** and **amount**. The amount bar goes both ways: drag left for negative. Set the source to `-` to turn a slot off. `Rensa` clears all slots.

`LFO 1-4`	The four LFOs.
`Volym-env, Env 2, Env 3`	The three envelopes.
`Velocity, Ton, Modhjul`	How hard you play, which note, and the mod wheel.
`Makro 1-4`	The four macro knobs in the top bar.
`Slump`	A new random value on every note.
`Rytm 1-3`	The euclidean rhythm lanes.

11 ENVELOPES, LFOS AND EFFECTS

ENVELOPES

Volym is the volume envelope. **Env 2** is wired to the filter through the knob `Env 2`. **Env 3** is free to use in the matrix. `Mono` plays one note at a time, and `Glid` slides between notes.

LFOS

Shapes: Sinus, Triangel, Såg, Fyrkant, S&H, **Mjuk** (smooth random), **Studs** (a bouncing ball), **Kaos** (chaotic) and **Euklid** (euclidean pulses, set with Slag and Steg). With `Synk` on, the LFO follows the tempo, from 1/16 up to 16 bars. `Omstart` restarts it on every note. `Mjukhet` smooths the edges.

EFFECTS

`Dist`, `Chorus`, `Eko` (a ping-pong echo in time) and `Rum` (room). The echo and the room duck while you play and come up in the gaps, so the sound stays clear. The output has a soft ceiling.

12 SPÄNNARMERING: THE SEQUENCER

New in ARMERING 5. A step sequencer built for minimal and percussive techno. Turn on `SEKVENS` and press play in your DAW (or `KLOCKA`). Every sound comes with its own pattern. Keys you play transpose the sequence: C3 plays it as written.

`A B C D`	Four variants, up to 64 steps each. The change happens on the next bar, so you can switch live.
`Step row`	Click or drag to turn steps on and off. The white frame shows where it plays.
`Lanes`	TON (note in the scale), STYRKA (velocity), CHANS (probability), RATCHET (1–4 hits in one step), LÄNGD (note length).
`Locks`	FILTER, POSITION, DECAY, DRIVE and KROPP: each step can set its own value, Elektron style. Double-click a bar to remove the lock.
`Lock length`	The locks have their own length. 7 locks against 16 notes slide against each other: polymeter.
`DRIFT`	MINUS drags unevenly, 909 pushes the offbeats late, BÅTVARV lets the tempo wander slowly. Always late, never before the beat. The knob sets how much.
`MUTERA`	The pattern changes a little every 8–64 bars, always from the original, so it never drifts away.
`MIDI UT`	Sends the notes as MIDI, so you can record them or play another instrument.

13 KROPP, KICK AND ÅLDER

`KROPP` plays the synth through a resonant body: **BETONG** (dense and dull), **METALL** (bright and long), **TRÄ** (round knock) or **GLAS** (high and ringing). MIX, KLANG (how long it rings) and TON (size).

`KICK`: send your kick to ARMERING's sidechain input (in Bitwig: the sidechain selector at the top of the device). `DUCK` lowers the synth when the kick hits, `FILTER` opens or closes the filter, `BÖJ` bends the pitch down. The kick is also a source in the mod matrix (`Kick`). These settings stay when you change sound.

`ÅLDER` ages the circuit: every voice gets its own tuning, filter and timing, like old components. At zero, ARMERING sounds exactly like before.

14 RECIPES

POLYRHYTHM ON THE FILTER

1. Load a BAS sound. Set `Rytm 1` to 16 steps and 4 hits, `Rytm 2` to 7 steps and 3 hits.
2. Drag both tiles onto `Cutoff`. Lower the cutoff a bit.
3. Turn `MAGI` up to about a third.

A PAD THAT BREATHES

1. Load a MATTA sound. Set LFO 2 to `Mjuk`, 8 bars.
2. Drag `LFO 2` onto `Position` in both oscillators.
3. Add a little `Rum`.

BOUNCING METAL

1. Load a KLONK sound. Set an LFO to `Studs` and turn on `Omstart`.
2. Drag it onto `Halvton` in OSC B, with a small amount.

ONE NOTE, A WHOLE SEQUENCE

1. Load a SEKVENS sound and hold one low note.
2. Press `SPARKA` until the movement grabs you. Record it.

15 NEW IN 5.1: THE COMPACT WINDOW

ARMERING 5.1 fits on a laptop screen. Same sound, same controls, a tighter window (1280 × 700).

The top half (oscillators, filter, FILTERRYTM) is unchanged. The lower panels now sit in three tabs:

TAB	WHAT IS IN IT
`SEKVENS · KROPP`	The SPÄNNARMERING sequencer (steps, lanes, locks, variants A–D, pages) next to KROPP, KICK and ÅLDER.
`RÖRELSE`	LFOs and the movement section.
`MATRIS · ENV · FX`	The modulation matrix, the envelopes and the effects.

Projects made with ARMERING 4 or 5.0 open exactly as before. Nothing in the sound has changed.

16 TROUBLESHOOTING

ARMERING does not show up	Quit the DAW completely and start it again, or rescan plug-ins. On Mac, run the installer again: it removes macOS's quarantine on the plug-in. On Windows, check that `ARMERING.vst3` is in C:\Program Files\Common Files\VST3.
macOS says the plug-in is damaged	That is macOS's block on downloaded files. Run `bash ~/Downloads/ARMERING/installera.command` again.
Windows protected your PC	Click **More info** → **Run anyway**. The plug-ins are not signed with a paid certificate.
The key does not work	Paste the whole key, including the dashes. A key for one machine only unlocks that machine; a bundle key unlocks all.
No sound	Press play in the DAW: the machines follow the transport. Instruments go on an instrument track, effects on an audio track or bus.

Anything else? Ask through the store, and the person who built the machine answers.

MASKINHALLEN